~~MNP~~ ~~CSE~~

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<u>Identity</u> = There exist a fix element e_V:

$$e_V * a = a * e_V = a.$$

<u>Inverse</u>: For each element $a \in V$ there exist a unique element.

a' (inverse element) : $a' * a = a * a' = e_V$

$(\mathbb{R}, \times)$

identity

$e_{\mathbb{R}} = I_{\mathbb{R}}$

$2 \times \left(\frac{1}{2}\right) = 1_{\mathbb{R}}$

$(\mathbb{R}, +)$

$e_{\mathbb{R}} = 0_{\mathbb{R}}$

$2 + (-2) = 0_{\mathbb{R}}$

Suppose,

$a, b \in \mathbb{R}$, and define $\boxed{a * b := a + b - 1}$

identity & Inverse.

<u>fix (e)</u> <u>Identity</u>

L.S $e * a = a$ | And R.S $a * e = a$

$\Rightarrow e + a - 1 = a$ | $\Rightarrow a + e - 1 = a$

$\Rightarrow e = 1$ | $\Rightarrow e = 1$.

Because both identity are same hence $e = 1$

<u>Inverse</u>

pick any $a \in \mathbb{R}$ then assume a has a inverse and denoted it by a'.

$a' * a = \underline{e}$ → Its necessary to know the value of identity.

$\Rightarrow a' + a - 1 = 1$

$\Rightarrow a' = 2 - a$.

Vector Space

Physisist :

CSE: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Math: 10 Axioms $\rightarrow$ Precondition

(set, $*$, $\odot$)

$\boxed{\ } \odot v$ Fields α, β, γ

$\odot$

$* = +$

$a + b \in V$

$a + b = b + a$

$(a+b)+c = a+(b+c)$

0

$(-a)$

(i) Closure : $\alpha \in F = \{\mathbb{R}, \mathbb{C}\}$

$\alpha \odot v \in V$

(ii) $\alpha \odot (u+v) = \alpha \odot u + \alpha \odot v$.

(iii) $(\alpha + \beta) \odot u = \alpha \odot u + \beta \odot u$

~~(iv) $(\alpha\beta) \odot \vec{u}$~~

(iv) $\beta \odot (\alpha \odot u) = (\beta * \alpha) \odot u$

(v) Identity,

I_v,

Any $\alpha \in V$, then $I_v \odot \alpha = \alpha \odot I_v = \alpha$

Verify vector space

$V = \mathbb{R}^2$

Set

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 y_2 \end{pmatrix}$$

$$k \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} k x_1 \\ k y_1 \end{pmatrix}$$

verify Is $(V, +, \odot)$ a vector space or not?

(A 1) is satisfied

(A2) is satisfied

(A3) $u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ $v = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ $w = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$

(A4) Suppose, $e_{\mathbb{R}^2} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$e_{\mathbb{R}^2} + u = u$ $\qquad$ $u + e_{\mathbb{R}^2} = u$

$$\Rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha + x_1 \\ \beta y_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$\alpha + x_1 = x_1$ and $\beta y_1 = y_1$

$\Rightarrow \alpha = 0 \quad \beta = 1.$

$\therefore e_{\mathbb{R}^2} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$

(A5) <u>Inverse:</u> $u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ then assume the inverse element of u is $u' = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$u' + u = e_{\mathbb{R}^2}$ from (A4) $\qquad$ $u + u' = e_{\mathbb{R}^2}$

$$\Rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha + x_1 \\ \beta y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\alpha + x_1 = 0 \qquad \beta y_1 = 1$$

$$\Rightarrow \alpha = -x_1 \quad \Big| \quad \beta = \frac{1}{y_1}$$

$$u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \Rightarrow u' = \begin{pmatrix} -x_1 \\ 1/y_1 \end{pmatrix} \rightarrow y_1 \neq 0$$

But it can't be.

For each element $u \in V$ there exist an unique element u' such that $u' * u = u * u' = e$ ✓.

Example 2

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

$$K \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} Kx_1 \\ y_1 \end{pmatrix}$$

Checklist

Addition

A3 : Associativity (Thus A1 and A2 is checked)

A4 + A5 : Identity + Inverse

A2 : commutativity

A1 : Closure

Multiplication

M2 : Vector Addition

M4 : Compatibility

M3 : Field Addition

M5 : Identity

M1 : Closure